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Functional Performance Benchmarking for Organic Molecules

1.0 Introduction and Background

The development of new pharmaceuticals is a time-consuming, and expensive process. On average, the development of a drug formula that is ready for human testing and marketing takes approximately 7 years [1]. To minimize development time, companies can utilize density functional theory (DFT) computational simulations to their advantage by using it as a pre-screening tool to analyze properties of molecular structures to determine a jumping off point for physical synthesis. DFT is a quantum-mechanical method that is used to calculate the electronic structure of atoms, molecules, and solids [2]. Furthermore, computational speed and accuracy can be tuned to the application by the choice of functionals and basis sets used, further enhancing predictive capabilities. Functionals are mathematical models that are used to map the electron density of a system to its total energy. Basis sets are mathematical functions used to form molecular orbitals to describe electron density to predict characteristic geometries, energies, and reaction properties.

The overall motivation of this project is to assess which functional would be best suited for high-throughput analysis of organic molecules to reduce in-lab troubleshooting of pro-drug compound synthesis. The following five functionals will be studied: LDA, PBE, B3LYP, ωB97X-D3, and M06-2X. The functionals increase in accuracy and computational cost, respectively. LDA calculates energy by making the assumption that variations in electron density are slow, and treats the local density as a uniform electron gas [3]. PBE uses the generalized gradient approximation (GGA) to calculate electron-exchange correlation based on the density and its gradient [4]. B3LYP is a hybridized functional that combines electron exchange and local and gradient-corrected correlation functionals with empirically determined constants [5]. ωB97X-D3 is a range-separated functional that combines short- and long-range exchange with empirical dispersion corrections to improve accuracy [6]. M06-2X is a meta-hybrid functional with a high percentage of Hartree-Frock (HF) exchange designed to improve accuracy for thermochemistry, kinetics, and interaction energy calculations [7]. By comparing the performance of different functionals against experimental data, the accuracy of the computed geometries and reaction energies can be evaluated. Furthermore, the computational cost of each functional and how each functional handles computations for molecules of different sizes will be evaluated. Twenty molecule structures will be investigated to provide a comprehensive analysis of the performance of each functional. To determine which functional would be best for high-throughput analysis of large molecules, a low cost to high accuracy ratio would be required.

2.0 Methodology

To efficiently benchmark the selected molecules, an automated computational pipeline was developed to execute over 100 independent calculations. The ORCA quantum chemistry software was utilized as the primary simulation engine to compute molecular properties. ORCA operates using plain-text input files ".inp" that combine operational keywords with Cartesian coordinates formatted as "XYZ" files. To streamline the dispatch of these calculations and automatically parse the resulting data, a custom Python interfacing package, named Nano Chemistry Library (NCL) was integrated into the workflow [25]. Prior to calculation, the Avogadro 2 software suite was used to visualize the initial molecular structures and convert literature coordinate files from the ".SDF" format into the required ".XYZ" format.

The complete benchmarking pipeline consisted of several sequential steps. First, initial geometrical properties were established from literature sources. Next, geometric optimizations were performed on each molecule across all evaluated functionals to extract the total molecular enthalpy. Subsequently, single-point calculations were executed for every constituent atom across all functionals to extract the final single point energies. These calculated values were used to compute the molecules enthalpy of formation (ΔH_f), which is the parameter used to quantify the computational accuracy. This was achieved by subtracting the total computed enthalpy of the individual atoms from the enthalpy of the optimized molecule. Finally, these computed enthalpies of formation were compared against experimental literature values, allowing the Mean Absolute Error (MAE) to be calculated to assess the predictive fidelity of each functional.

The benchmarking dataset was categorized into three distinct chemical groups: alkanes, organics, and pro-drugs. Alkanes were selected for their structural simplicity and the predictable scaling of their properties with increasing carbon chain length. The organic category was included to assess the computational effects of introducing heteroatoms (such as nitrogen, oxygen, and fluorine) alongside structural features like alcohol groups and cyclic rings. Finally, the pro-drug category was incorporated to evaluate whether the selected functionals could accurately model the larger, complex intermediates representative of pharmaceutical development. A comprehensive list of the target molecules for each category is detailed in *Table A1*. To generate the initial computational models, reference structures were sourced from the PubChem database in the ".SDF" format. These reference files were subsequently visualized and converted into the required ".XYZ" format using the Avogadro 2 software, establishing a structural baseline to later evaluate the accuracy of the computationally optimized geometries.

Following the generation of the initial structural files, the geometries of all selected molecules were computationally optimized to their respective energetic ground states utilizing the ORCA software suite. This structural optimization was driven by the specific density functional being evaluated in conjunction with the "DEF2-SVP" basis set, which was applied uniformly across all computational trials. The geometric relaxation sequence was initiated using the standard "OPT" keyword. However, because different density functionals map electron topography uniquely, computational optimizations can frequently converge onto local energy maxima or transition states, commonly referred to as saddle points, rather than the true global energetic minimum.

To rigorously verify that a stable ground-state geometry was achieved, a vibrational frequency "FREQ" calculation was systematically executed immediately following the completion of each optimization cycle. These computed vibrational frequencies serve as a diagnostic tool for structural stability. If all resulting frequencies are strictly positive, the molecular geometry is confirmed to be fully optimized. Conversely, the presence of one or more imaginary (negative) frequencies indicates an unstable saddle point, necessitating further structural refinement. To resolve these transition states, the molecular geometry was deliberately distorted by displacing the atomic coordinates along the specific normal mode associated with the imaginary frequency. This structural displacement was achieved through a matrix multiplication between the currently optimized atomic coordinates and the column vector corresponding to the problematic normal mode. This iterative cycle of optimization and coordinate displacement was repeated until all imaginary frequencies were eliminated, yielding a fully optimized molecule.

To manage this recursive and computationally demanding process, the custom Nano Chemistry Library (NCL) was employed as the automated interface between the Python scripting environment and the ORCA simulation engine [25]. Within the NCL framework, the "Molecule" class was utilized to parse and load the ".XYZ" coordinate files directly into system memory. Subsequently, the "GeoOpt" class was deployed to autonomously dispatch the paired "OPT" and "FREQ" calculations to an isolated Docker container, ensuring a consistent and reproducible computational environment. The "GeoOpt" class

algorithmically governed the entire iterative optimization loop, automatically recalculating and applying normal mode displacements until a verified ground state was achieved for every molecule. A comprehensive visual representation of this automated optimization workflow is provided in *Figure 1*.

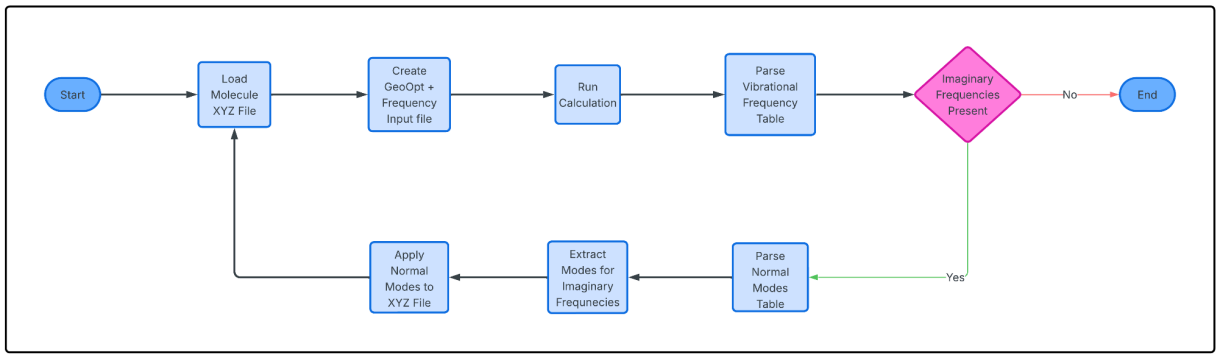

Figure 1. GeoOpt class Flowchart

Following the successful geometric optimization of the molecules, the standard enthalpy of each constituent atom was determined to facilitate the calculation of the molecular enthalpy of formation. As defined by *Equation 1*, computing the individual atomic enthalpy primarily required the final single-point electronic energy (E_{elec}). This energy value was acquired through a single-point calculation in ORCA, utilizing strictly the "method/basis" and "functional" keywords. These atomic calculations were systematically dispatched across all evaluated functionals using the NCL automated workflow with the same basis set and functionals. To complete the enthalpy calculation, the thermal correction parameter, $\frac{5}{2}RT$, was derived using the Ideal Gas Constant (R) and the standard computational temperature (T) of 298.15 K. Furthermore, the atomic experimental enthalpy correction term ($H_{exp,f,corr}$) was incorporated based on established literature values, which are summarized in *Table A6*. Ultimately, these computed individual atomic enthalpies were utilized in conjunction with the optimized molecular enthalpy to calculate the final enthalpy of formation via *Equation 2*.

$$H_{atom} = E_{elec} + \frac{5}{2}RT + H_{exp,f,corr}$$

Equation 1. Standard Enthalpy of an Atom

$$\Delta H_f = H_{molecule} - (\sum H_{atom})$$

Equation 2. Enthalpy of Formation for a Molecule

In the final stage of the computational pipeline, the data generated from each calculation was systematically parsed using the NCL library [25]. Specifically, the custom "OrcaOutputFile" class was employed to automatically access and read the ".OUT" files, extracting both the total molecular enthalpies and the final single-point atomic energies. These extracted thermodynamic values were then organized into dedicated data tables categorized by their respective density functionals. All parsed raw data, subsequent thermochemical calculations, and the corresponding error analyses have been comprehensively compiled and presented in the *Results and Figures* section of this report.

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3.0 Results and Figures

To analyze the performance of each functional, the Mean Absolute Error of enthalpy of formation will be evaluated. Furthermore, the bond lengths and angles of each structure will be compared to optimized geometries from literature. Finally, an analysis of the computational cost versus accuracy will be conducted. The results can be seen in the below figures and tables, as well as in the appendix.

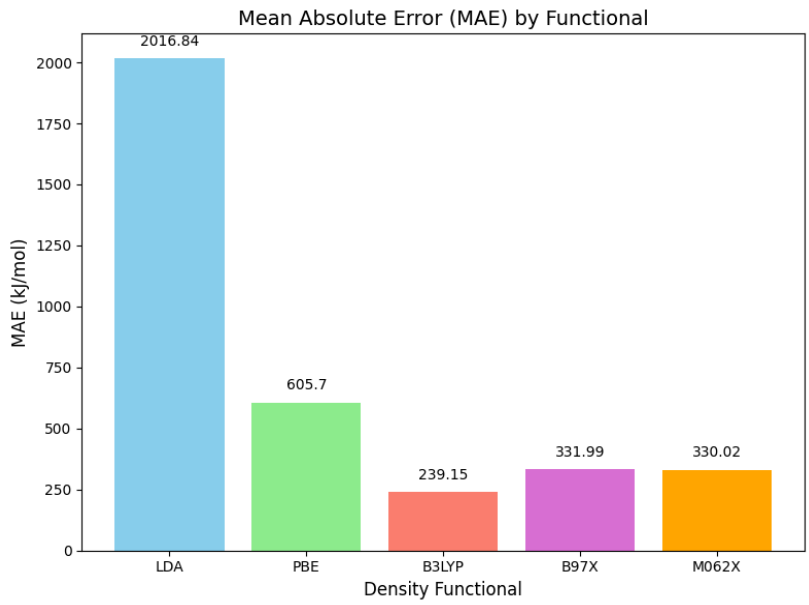

Figure 2. Mean Absolute Error of enthalpy of formation for each functional

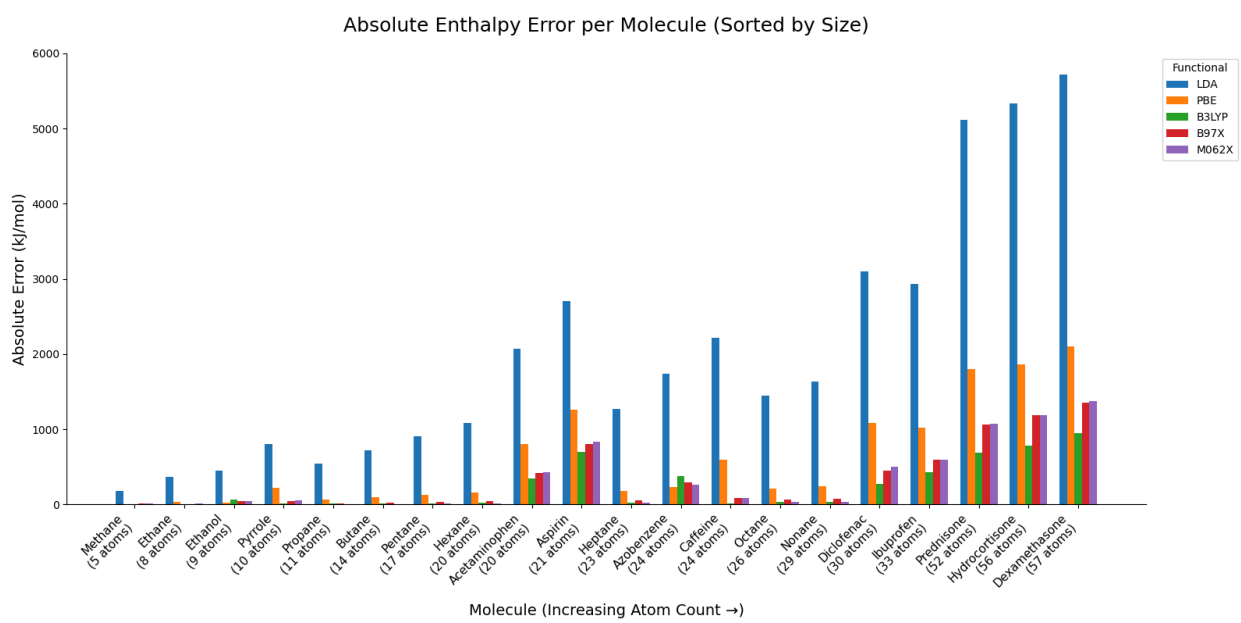

Figure 3. Absolute enthalpy of formation error per molecule

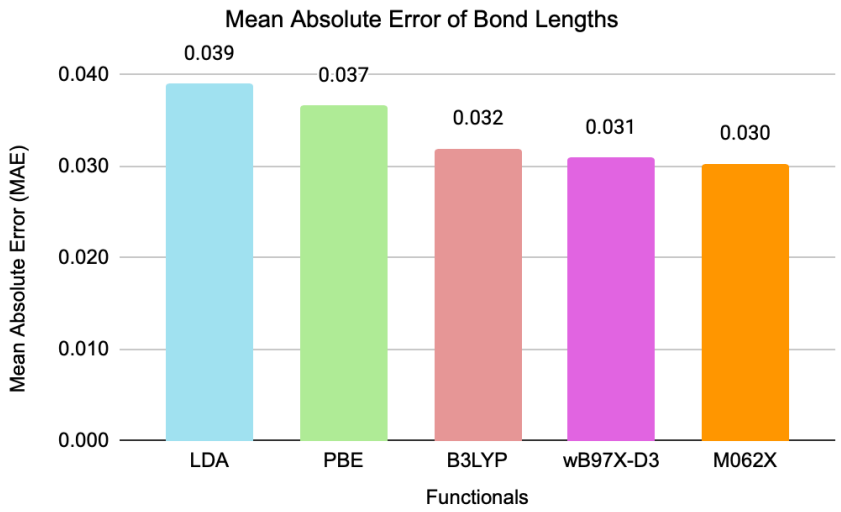

Figure 4. Mean Absolute Error of bond lengths (Å)

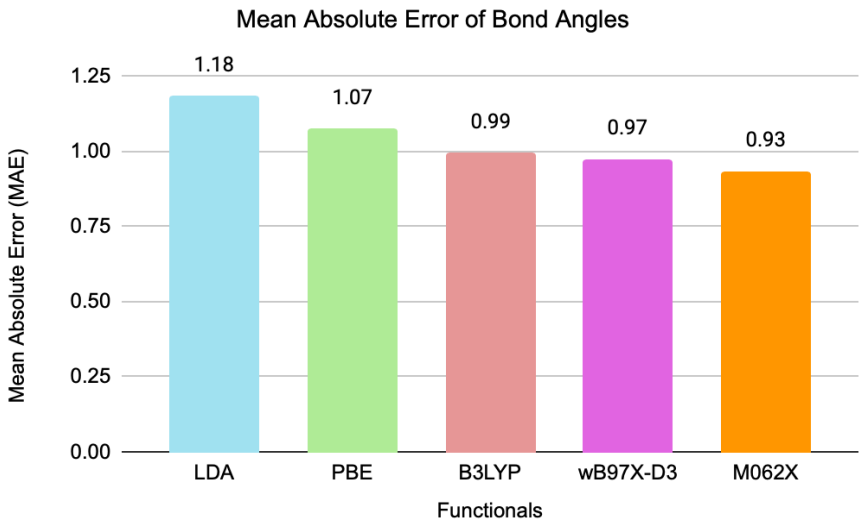

Figure 5. Mean Absolute Error of bond angles (°)

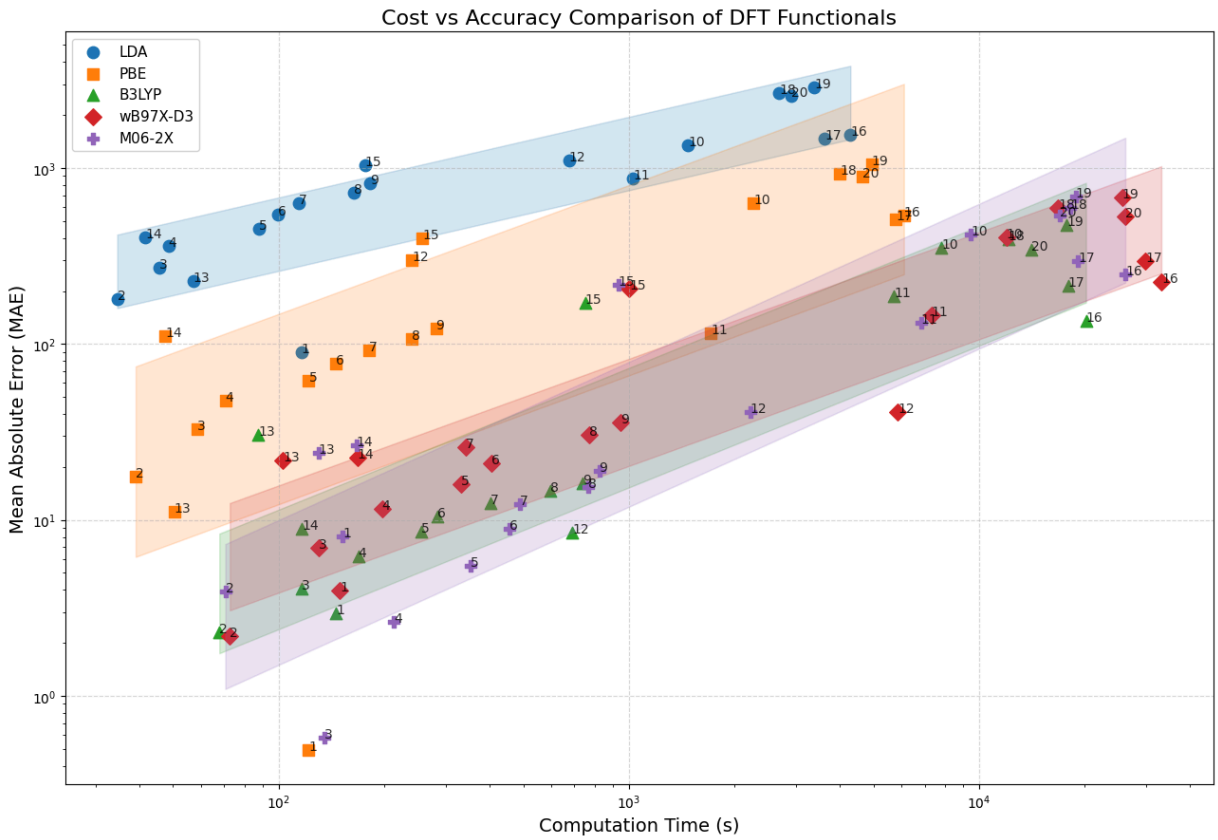

Figure 6. Computational cost analysis. Index of each molecule detailed in *Table A1*.

4.0 Discussion and Physical Interpretation

An analysis of the generated figures and data tables reveals a clear inverse relationship between the computational complexity of the selected density functional and the resulting Mean Absolute Error (MAE). As illustrated in *Figure 2*, the "B3LYP" functional emerges as the most optimal model for predicting the enthalpy of formation, closely followed by the more complex "ωB97X-D3" and "M06-2X" functionals. While the inclusion of dispersion corrections and meta-hybrid parameters in the larger functionals did not strictly outperform “B3LYP” for thermochemical enthalpy, all hybrid models delivered an order of magnitude higher accuracy compared to the baseline "LDA" functional. Physically, this discrepancy is expected, the “LDA” functional treats electron density as a uniform gas, an approximation that fails to capture the localized and heterogeneous electron distributions in covalent organic bonds. The incorporation of exact Hartree-Fock exchange in “B3LYP” drastically corrects for these localized electron-electron interactions, resulting in the sharp reduction in MAE.

When observing the error as a function of molecular size, as arranged in *Figure 3*, the MAE generally scales proportionally with the number of atoms. However, significant outliers deviate from this linear trend. Molecules such as Acetaminophen, Aspirin, and notably Diclofenac exhibit disproportionately high computational errors relative to their atomic counts when compared to simple alkanes like Octane or Nonane. This deviation can be physically attributed to their complex structures, specifically the cyclic aromatic benzene rings coupled with highly electronegative atoms such as Nitrogen, Oxygen, and Chlorine. The presence of these atoms, and the associated double bonds in the benzene ring create dense, highly correlated electron environments. As mentioned in the previous

paragraph, “LDA” fails to capture the localized electron interactions, and similarly, the simpler functionals struggle to map these non-covalent, intra-molecular interactions and charge distributions accurately.

A similar trend of diminishing returns is observed in the geometrical accuracy of the simulations. Examining *Figure 4* and *Figure 5*, the MAE for both bond lengths and bond angles decreases sequentially across the "LDA", "PBE", and "B3LYP" functionals before noticeably plateauing for "ωB97X-D3" and "M06-2X". This plateau indicates that the spatial resolution of the chosen "DEF2-SVP" basis set has been reached. Once a density functional adequately accounts for electron exchange and correlation, the geometric precision of the molecular orbitals becomes bottlenecked by the limited number of mathematical functions provided by a double-zeta basis set. To extract the full predictive capabilities of the advanced functionals and further reduce geometrical MAE, a larger basis set incorporating diffuse and higher-polarization functions would be required.

Finally, an evaluation of computational cost versus accuracy, depicted in *Figure 6*, highlights the practical trade-offs of density functional selection for high-throughput screening. The "LDA" and "ωB97X-D3" functionals demonstrate the most consistent trends between cost and accuracy, albeit at opposite ends of the computational spectrum. Conversely, the "PBE" functional displays the highest variance, providing highly accurate results for certain structures while suffering from severe predictive failures for others. Furthermore, *Table A5* reveals extreme computational time outliers among the pro-drug category. For instance, the optimization of Diclofenac required over 33,000 seconds using the "ωB97X-D3" functional, a cost severely disproportionate to simpler organic molecules of similar atomic size. This extreme computational bottleneck is driven by the combination of aromatic flexibility and the presence of heavier halogen atoms like Chlorine in Diclofenac or Fluorine in Dexamethasone. Heavier atoms possess significantly more core electrons, exponentially increasing the number of two-electron integrals that must be computed per self-consistent field (SCF) cycle, and severely complicating the resolution of imaginary frequencies and saddle points during the geometric optimization process.

Additionally, the regression slopes depicted in *Figure A2* provide further insight into the relationship between computational cost and predictive error across the tested functionals. The steeper regression slopes observed in the hybrid and meta-hybrid functionals suggests that computational expense scales more drastically with increasing molecular complexity. This suggests that the “ωB97X-D3” and “M06-2X” functionals may be limited in their practicality for larger molecular systems and high-throughput screening due to greater computational demands. This trend is further proven in *Figure A1*, demonstrating the extreme computational times for the molecules belonging in the pro-drug category due to the computational bottleneck mentioned previously. Furthermore, the regression analysis reveals that the “B3LYP” and “ωB97X-D3” functionals demonstrate the strongest correlation of cost to accuracy, with R^2 values exceeding 0.8. A strong correlation indicates greater predictability when scaling molecular systems from consistent performance trends across the molecule sample set. In contrast, PBE has the lowest R^2 value, suggesting greater variance in performance and reduced reliability when predicting molecular properties. As a result, utilizing the “PBE” functional would not be suitable for chemically diverse systems. Overall, the “B3LYP” functional would be the most practical option to screen a chemically diverse dataset due to its predictive reliability and lower computational cost.

For a comprehensive overview of all computational results, the remaining data and tables can be found in the *Appendix*. This supplementary section contains the complete compilation of raw single-point energies, geometric parameters, extended thermochemical tables, and additional figures for all twenty molecules across the evaluated functionals, further supporting the trends and physical interpretations discussed above.

5.0 Conclusion

This study aimed to optimize the pre-screening phase of pharmaceutical development, a process that traditionally requires years of extensive research, by evaluating the efficacy of Density Functional Theory (DFT) for high-throughput molecular analysis. An automated computational pipeline, driven by the ORCA simulation engine and the custom Nano Chemistry Library (NCL), was successfully established to benchmark twenty molecules across three distinct categories: alkanes, organics, and pro-drugs. The results demonstrated a clear inverse relationship between density functional complexity and Mean Absolute Error (MAE). Hybrid functionals, particularly "B3LYP", successfully balanced computational cost with high predictive accuracy for the enthalpy of formation, drastically outperforming baseline models by effectively correcting for localized electron-electron interactions.

Despite these successes, significant computational limitations and failures were identified. The simpler "LDA" and "PBE" functionals failed to capture non-covalent intra-molecular interactions, leading to severe predictive inaccuracies for complex pro-drugs containing aromatic rings and highly electronegative heteroatoms. Additionally, a clear plateau in geometrical accuracy (specifically concerning bond lengths and angles) was observed across the more advanced functionals, indicating that predictive capabilities were bottlenecked by the limited spatial resolution of the "DEF2-SVP" double-zeta basis set. Finally, extreme computational bottlenecks emerged when optimizing large, flexible pro-drug structures containing heavy halogen atoms, for example, the optimization of Diclofenac using the "ωB97X-D3" functional required a disproportionate amount of computational time compared to simpler molecules.

To address these limitations and further refine the high-throughput screening pipeline, future work must explore a wider array of computational parameters. Subsequent studies should incorporate a broader selection of density functionals, specifically testing higher-order functionals to determine if thermochemical and geometrical predictive accuracy can be pushed even further. Furthermore, transitioning to a more complex basis set, specifically those incorporating diffuse and higher-polarization functions, is necessary to overcome the observed plateau in geometric precision. Finally, a comprehensive comparative analysis of basis sets should be conducted to identify the most efficient configurations tailored to specific chemical groups. This analysis will definitively establish which basis sets are best suited for the rapid screening of simple alkanes, versus the more demanding computational requirements of functionalized organics and large pro-drug intermediates.

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7.0 Appendix

Table A1. Molecules belonging to each category and their index

Alkane Molecules		Organic Molecules		Pro-Drug Molecules	
1	Methane	10	Aspirin	15	Acetaminophen
2	Ethane	11	Azobenzene	16	Diclofenac
3	Propane	12	Caffeine	17	Ibuprofen
4	Butane	13	Ethanol	18	Hydrocortisone
5	Pentane	14	Pyrrole	19	Dexamethasone
6	Hexane		-	20	Prednisone
7	Heptane		-		-
8	Octane		-		-
9	Nonane		-		-

Table A2. Mean Absolute Error of Enthalpy of Formation

Molecule	LDA (kJ/mol)	PBE (kJ/mol)	B3LYP (kJ/mol)	wB97x-D3 (kJ/mol)	M06-2X (kJ/mol)
Methane	90.192	0.492	2.948	3.985	8.054
Ethane	180.806	17.591	2.297	2.180	3.938
Propane	271.529	32.740	4.083	6.931	0.578
Butane	362.112	47.577	6.194	11.529	2.641
Pentane	452.398	62.157	8.600	16.001	5.499
Hexane	543.192	77.244	10.447	20.967	8.903
Heptane	633.863	92.196	12.483	25.836	12.223
Octane	724.301	106.927	14.686	30.460	15.284
Nonane	815.447	122.368	16.219	35.805	19.081
Aspirin	1352.725	629.071	351.306	402.806	418.992
Azobenzene	871.011	114.762	186.230	146.042	131.917
Caffeine	1106.678	298.234	8.504	40.948	41.118
Ethanol	227.181	11.181	30.420	21.769	23.935
Pyrrole	403.112	111.196	8.857	22.647	26.435

Acetaminophen	1037.417	400.169	170.293	207.234	216.003
Diclofenac	1550.492	540.029	134.507	224.667	249.254
Ibuprofen	1465.347	508.510	213.981	296.146	296.304
Hydrocortisone	2666.802	932.633	392.357	593.930	595.545
Dexamethasone	2857.531	1052.448	472.238	677.894	686.479
Prednisone	2556.278	899.509	344.872	532.123	538.017

Table A3. Mean Absolute Error of Bond Lengths

Molecule	LDA (Å)	PBE (Å)	B3LYP (Å)	wB97x-D3 (Å)	M06-2X (Å)
Methane	0.484382	0.483477	0.478047	0.475656	0.475430
Ethane	0.014847	0.015627	0.009753	0.007517	0.006349
Propane	0.015655	0.014927	0.009243	0.007039	0.005883
Butane	0.016531	0.014305	0.008720	0.006368	0.005332
Pentane	0.017078	0.013939	0.008439	0.006438	0.005592
Hexane	0.017466	0.013682	0.008225	0.006446	0.005675
Heptane	0.017908	0.013340	0.007913	0.006631	0.005897
Octane	0.018146	0.013165	0.007761	0.006649	0.005952
Nonane	0.018261	0.013142	0.007760	0.006649	0.005973
Aspirin	0.013125	0.011987	0.010240	0.010991	0.010444
Azobenzene	0.008716	0.013072	0.007813	0.006220	0.005684
Caffeine	0.016870	0.012417	0.009951	0.010590	0.010339
Ethanol	0.016943	0.013391	0.009346	0.009188	0.008062
Pyrrole	0.010333	0.012026	0.005818	0.004457	0.003584
Acetaminophen	0.011306	0.011834	0.008425	0.007838	0.007341
Diclofenac	0.012618	0.012574	0.008417	0.007350	0.006963
Ibuprofen	0.015003	0.012713	0.008610	0.007887	0.006610
Hydrocortisone	0.017992	0.012146	0.008047	0.008002	0.008145
Dexamethasone	0.017310	0.012239	0.007820	0.007986	0.007947
Prednisone	0.018017	0.012356	0.007999	0.007897	0.007900

Table A4. Mean Absolute Error of Bond Angles

Molecule	LDA (°)	PBE (°)	B3LYP (°)	wB97x-D3 (°)	M06-2X (°)
Methane	0.001528	0.001631	0.001904	0.002035	0.002334
Ethane	1.342694	1.158689	1.014809	1.001490	0.856990
Propane	0.777422	0.714268	0.647696	0.598374	0.482484
Butane	1.061063	0.943100	0.847525	0.830849	0.717209
Pentane	1.032287	0.940082	0.853366	0.814154	0.726919
Hexane	1.016610	0.941566	0.862544	0.821625	0.733838
Heptane	0.903203	0.814538	0.737163	0.701751	0.623410
Octane	0.896737	0.822236	0.747860	0.704241	0.618146
Nonane	0.911607	0.843584	0.774158	0.728642	0.649511
Aspirin	1.689926	1.610455	1.496514	1.458072	1.465543
Azobenzene	0.907988	0.803123	0.743040	0.729417	0.768909
Caffeine	1.434406	1.292592	1.208129	1.187132	1.218450
Ethanol	1.479348	1.468815	1.344173	1.325978	1.217320
Pyrrole	0.357753	0.254529	0.270174	0.277757	0.290934
Acetaminophen	1.578495	1.559817	1.503608	1.450205	1.439689
Diclofenac	1.904941	1.689956	1.608436	1.666300	1.682195
Ibuprofen	1.370741	1.352222	1.311906	1.251467	1.165099
Hydrocortisone	1.507734	1.323941	1.192485	1.195302	1.212460
Dexamethasone	1.673560	1.396164	1.336842	1.329435	1.367085
Prednisone	1.805288	1.560528	1.395037	1.381461	1.428508

Table A5. Computational time

Molecule	LDA (s)	PBE (s)	B3LYP (s)	wB97X-D3 (s)	M06-2X (s)
Methane	116.134	121.98	146.161	149.936	152.693
Ethane	34.636	39.123	67.817	72.626	70.547
Propane	45.671	58.461	116.538	130.405	135.036
Butane	48.518	70.523	169.132	197.519	213.483
Pentane	87.727	122.031	255.583	332.94	353.102
Hexane	99.416	145.79	283.862	405.75	457.54
Heptane	114.054	181.253	402.259	343.119	487.6
Octane	163.676	239.667	595.361	770.229	767.601
Nonane	182.655	281.773	737.505	949.183	823.65
Aspirin	1475.508	2274.076	7826.661	11994.91	9498.075
Azobenzene	1030.63	1716.27	5727.876	7324.949	6831.877
Caffeine	675.211	240.139	689.093	5863.862	2223.909
Ethanol	57.086	50.433	87.591	102.966	130.126
Pyrrole	41.64	47.55	116.544	168.106	167.182
Acetaminophen	176.43	257.567	750.769	1002.92	934.037
Diclofenac	4294.201	6106.613	20213.92	33175.89	26177.63
Ibuprofen	3626.889	5788.292	17994.38	29898.29	19179.09
Hydrocortisone	2687.865	4008.047	12124.35	16837.96	18233.23
Dexamethasone	3391.745	4942.042	17789.03	25660.98	18917.53
Prednisone	2905.239	4661.076	14138.37	26251.21	17030.28

Table A6. Atomic Experimental Enthalpy of Formation Correction

Atom	Exp Enthalpy Correction (Eh)	Exp Enthalpy Correction (kJ/mol)
Hydrogen	0.08303142259	217.999
Carbon	0.2729651495	716.67
Nitrogen	0.1800354218	472.683
Oxygen	0.09490497048	249.173
Fluorine	0.0302380499	79.39
Chlorine	0.04620148543	121.302

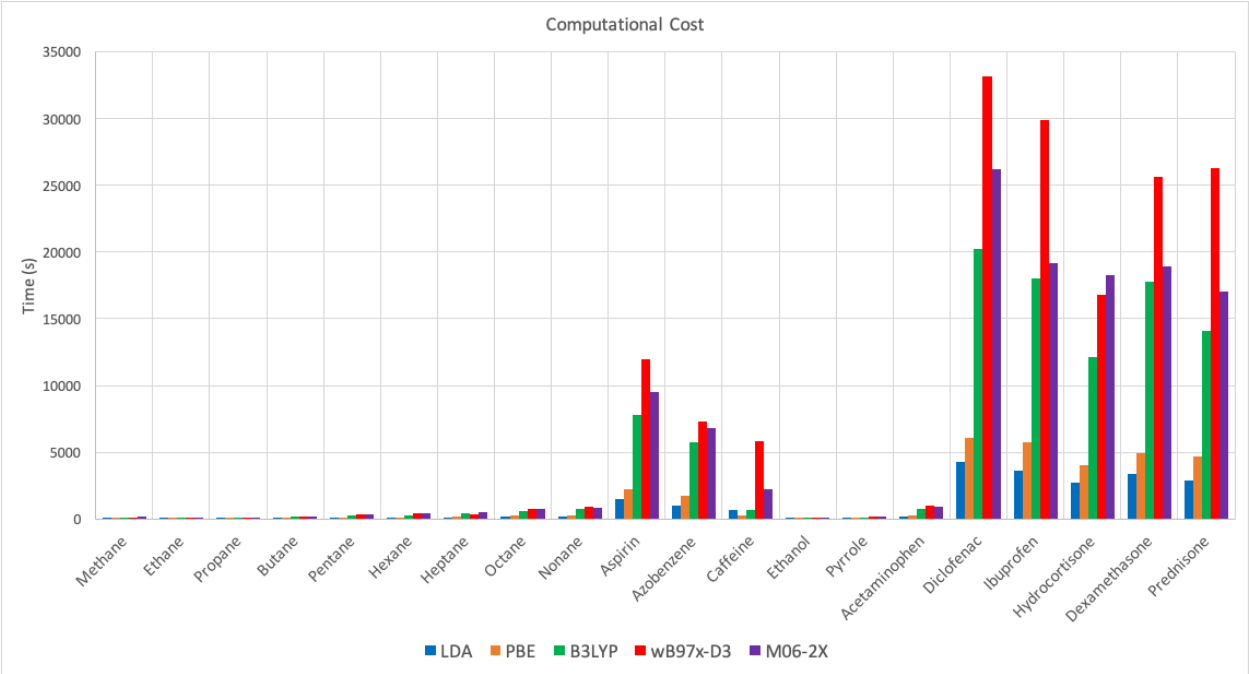

Figure A1. Computational cost for each molecule

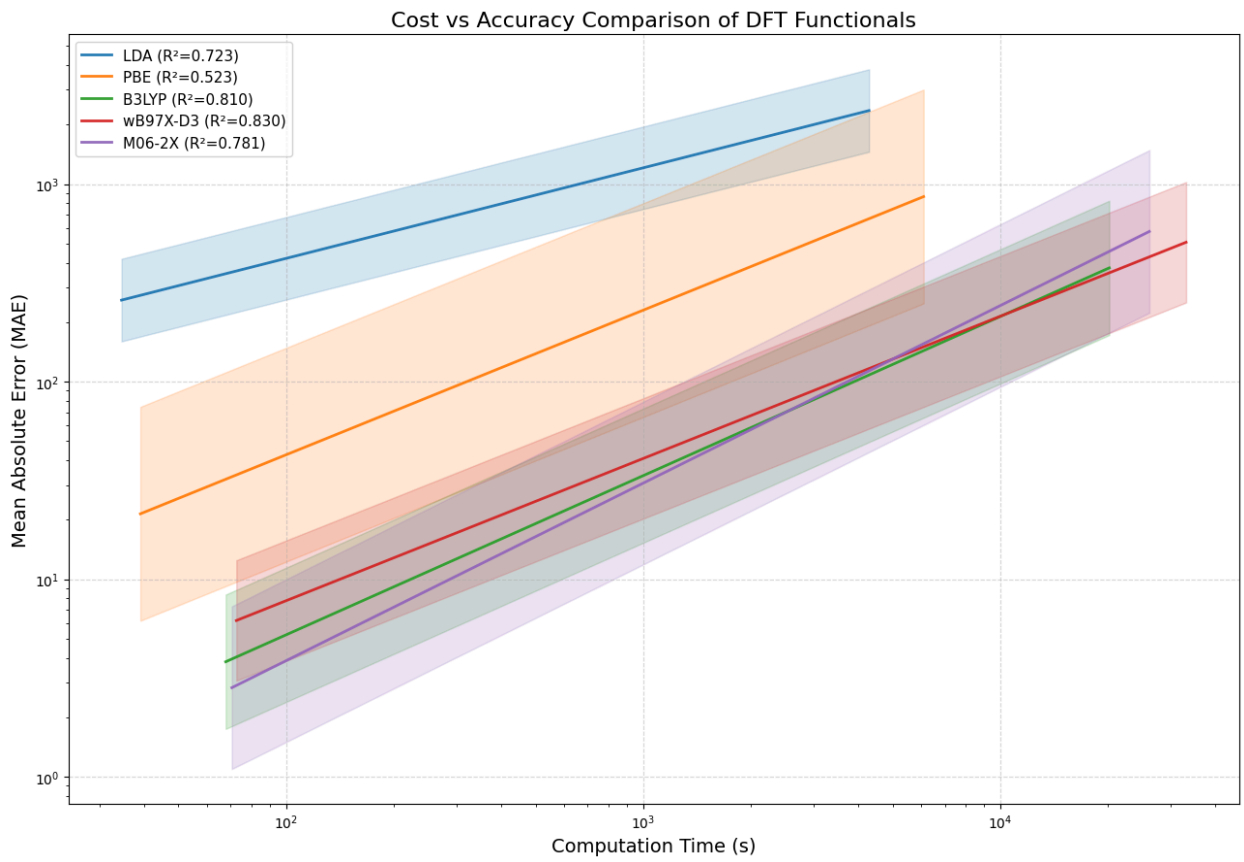

Figure A2. Linear best fit of computational cost analysis for each functional

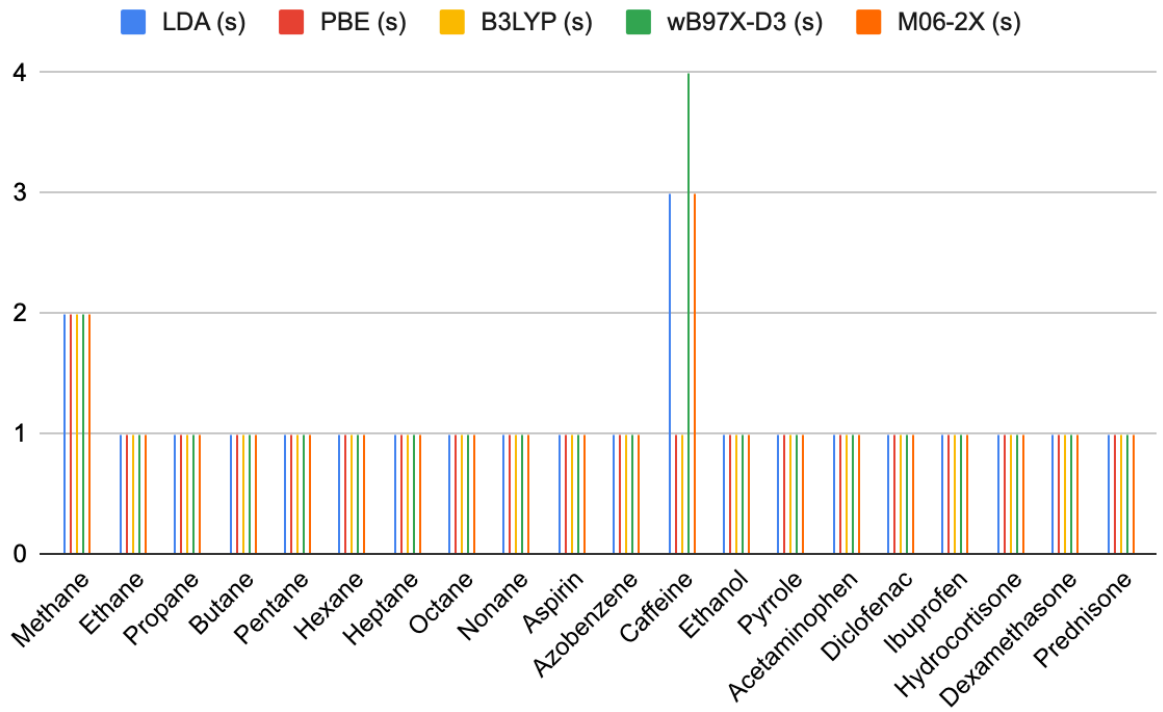

Figure A3. Computations required for each molecule

Table A7. Number of computations needed for each molecule

Molecule	LDA	PBE	B3LYP	wB97X-D3	M06-2X
Methane	2	2	2	2	2
Ethane	1	1	1	1	1
Propane	1	1	1	1	1
Butane	1	1	1	1	1
Pentane	1	1	1	1	1
Hexane	1	1	1	1	1
Heptane	1	1	1	1	1
Octane	1	1	1	1	1
Nonane	1	1	1	1	1
Aspirin	1	1	1	1	1
Azobenzene	1	1	1	1	1
Caffeine	3	1	1	4	3
Ethanol	1	1	1	1	1

Pyrrole	1	1	1	1	1
Acetaminophen	1	1	1	1	1
Diclofenac	1	1	1	1	1
Ibuprofen	1	1	1	1	1
Hydrocortisone	1	1	1	1	1
Dexamethasone	1	1	1	1	1
Prednisone	1	1	1	1	1